Welcome to module 3

Welcome back! Wow, you've learned a lot of new Python skills! You can use variables to store and label your data, and convert and combine different data types, such as integers and floats. You can call functions to perform useful actions on your data, and use operators to compare values. You also know how to write clean code that can be easily understood and reused by other data professionals. Finally, you can write conditional statements to tell the computer how to make decisions based on your instructions. Your knowledge of fundamental coding concepts is the first stage of a journey that leads to more advanced methods of data analysis.

And this learning journey will continue throughout your future career as a data professional. To me, that's one of the most exciting parts of the job! I'm always learning new ways to use Python for data analysis, whether from teammates at work, or from the super supportive online community. Learning new Python skills helps me become a more effective data professional. In this section of the course, you'll learn how to use Python code to automate repetitive tasks. Often data professionals need programs to perform an action repeatedly. For example, if you're analyzing sales data, you may want to perform the same calculation on hundreds of price values.

Rather than write new code each time you want to make the calculation, you can instead write an iterative statement, or loop. Loops automatically repeat a portion of code until a process is complete. We'll discuss two types of loops: while loops and for loops. Using Python to automate repetitive tasks saves me tons of time and effort, and reduces the risk of human error. It also reduces my overall workload and increases my productivity. I have more free time to focus on the main goal of any data analysis project: to generate insights for stakeholders. You can iterate over many different data types in Python, such as strings, lists, sets, and dictionaries.

In this section of the course, we'll focus on strings. Later on, we'll discuss the other data types in detail. As a data professional, you'll often work with strings when analyzing data. For example, you might examine textual data related to a company's products, services, customer feedback, and more. Operations like indexing and slicing allow you to select, filter, and edit data quickly and efficiently. These are valuable Python skills for any data professional. When you're ready to learn more, I'll meet you in the next video.